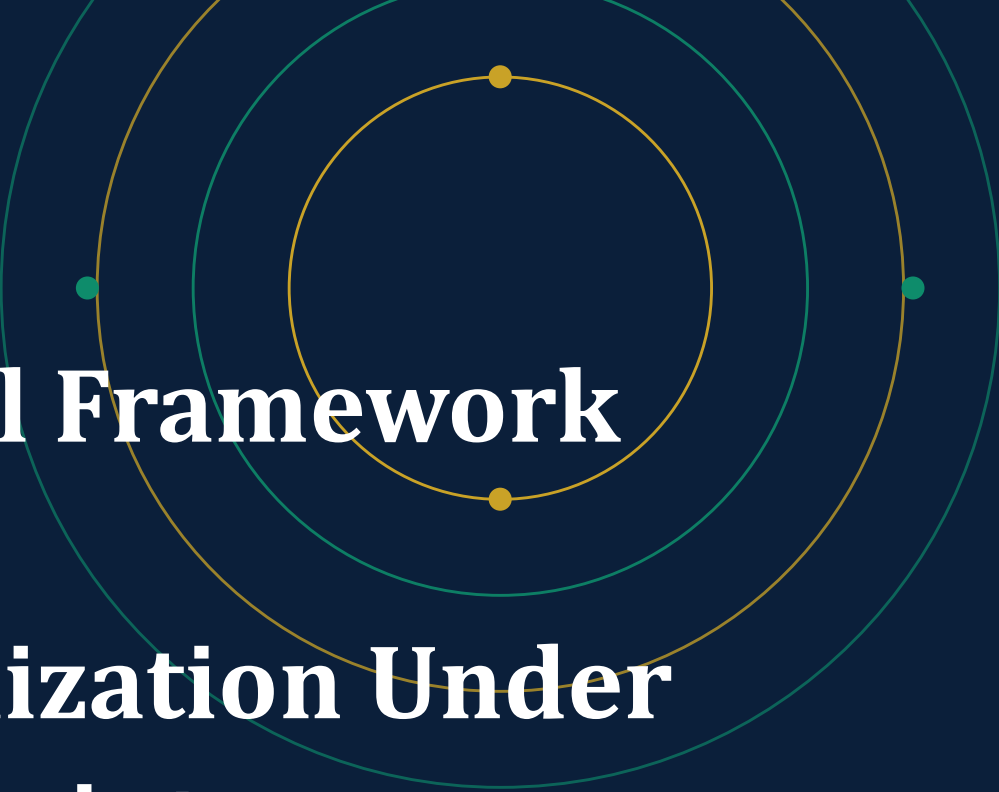




WISER GLOBAL QUANTUM + AI 2026 × VANGUARD CHALLENGE

A Hybrid Quantum–Classical Framework for Explainable Portfolio Optimization Under Real-World Financial Constraints

Multi-Asset Portfolio Construction & Call Center Staffing Optimization

QAOA Asset Selection • QUBO Formulation • Monte Carlo Risk Analysis • Explainable AI

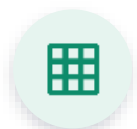
Pradeep Hariharanatham | Independent Research Submission

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THE PROBLEM

Real Portfolios Are Harder Than the Textbook

The Markowitz mean–variance model optimizes return against variance — but institutional mandates layer on discrete, combinatorial constraints the classical model was never built to hold.



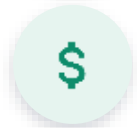
Cardinality Limits

Select exactly N assets from a candidate universe



Liquidity Requirements

Every holding must clear a minimum tradability bar



Transaction Costs

Trading frictions must be minimized, not ignored



Turnover & Sector Limits

Concentration and rebalancing rules bind the search

Every discrete "include / exclude" decision above is a binary variable — turning a smooth optimization into a mixed-integer quadratic program (MIQP).

WHY IT MATTERS

Institutional Reality

N

Discrete binary decisions per asset

MIQP

Problem class — computationally hard

↑↑

Search space grows combinatorially with universe size

As the candidate universe grows, classical solvers face a scalability wall precisely where institutional portfolios live.

Why Classical Optimization Falls Short

THE MARKOWITZ OBJECTIVE

minimize $w^T \Sigma w$

subject to: $\sum w_i = 1$ and $w_i \geq 0$

w = asset weights Σ = covariance matrix

Elegant for continuous allocation — but silent on which assets even qualify for the portfolio.



No native discreteness

Binary include/exclude decisions sit outside the convex QP formulation



Scalability wall

Combinatorial search space explodes as the candidate universe grows



Constraint patchwork

Cardinality & liquidity limits require ad-hoc heuristics, not principled solving

THE RESULT

A Mixed-Integer Quadratic Program (MIQP)

Real-world constraints transform the clean Markowitz quadratic program into a mixed-integer quadratic program — a problem class that is NP-hard in general and becomes intractable for classical exact solvers as the asset universe scales into the hundreds or thousands.

The Research Gap

Classical finance and quantum computing have each matured independently — but few frameworks unite them into a single, explainable, production-realistic pipeline.



Why Quantum Computing?

Asset selection is fundamentally a binary decision problem — exactly the structure quantum optimization algorithms are built to explore.



Binary-Native Structure

Each asset is a qubit: 1 = selected, 0 = excluded — a direct hardware-level mapping



Combinatorial Search

Superposition explores exponentially many candidate subsets in parallel



QUBO / Ising Fit

QAOA maps financial objectives directly onto a cost Hamiltonian



Present-Day Limits

NISQ hardware: restricted qubits, noise, and shallow circuit depth

Why a Hybrid Quantum–Classical Approach?

Split the problem by computational suitability: let quantum handle discrete selection, classical handle continuous allocation.



QUANTUM (QAOA)

Discrete Asset Selection

- Solves the QUBO-formulated selection problem
- Explores combinatorial subsets via superposition
- Outputs a feasible, constraint-satisfying subset



CLASSICAL (CONVEX OPT.)

Continuous Weight Allocation

- Solves mean–variance weighting on selected assets
- Efficient, mature, well-understood solvers
- Delivers the final capital allocation vector

Each method is deployed only where it is strongest — avoiding both NISQ hardware limits and classical scalability walls.

The QUBO Formulation

Every candidate asset becomes a binary variable x_i — 1 if selected, 0 if excluded. Four weighted terms encode the full investment objective.

QUBO OBJECTIVE FUNCTION

$$\min H(\mathbf{x}) = -\alpha \sum r_i x_i + \beta \sum \sum \sigma_{ij} x_i x_j + \gamma (\sum x_i - K)^2 + \delta \sum (1 - \ell_i + c_i) x_i$$



Return Term

$$-\alpha \sum r_i x_i$$

Rewards assets with higher expected return
 r_i



Risk Term

$$\beta \sum \sum \sigma_{ij} x_i x_j$$

Penalizes covariance — discourages
correlated, concentrated picks



Cardinality Term

$$\gamma (\sum x_i - K)^2$$

Pulls the selection toward exactly K assets



Liquidity / Cost Term

$$\delta \sum (1 - \ell_i + c_i) x_i$$

Discourages illiquid assets and high trading
costs

QAOA-Based Asset Selection

The QUBO objective is transformed into an Ising Hamiltonian, then solved by alternating quantum operators across variational layers.



$$|\psi(\gamma, \beta)\rangle = U(B, \beta_p)U(C, \gamma_p) \dots U(B, \beta_1)U(C, \gamma_1)|+\rangle^n \quad \text{— parameters re-optimized iteratively across } p \text{ variational layers}$$

The Five-Stage Hybrid Architecture

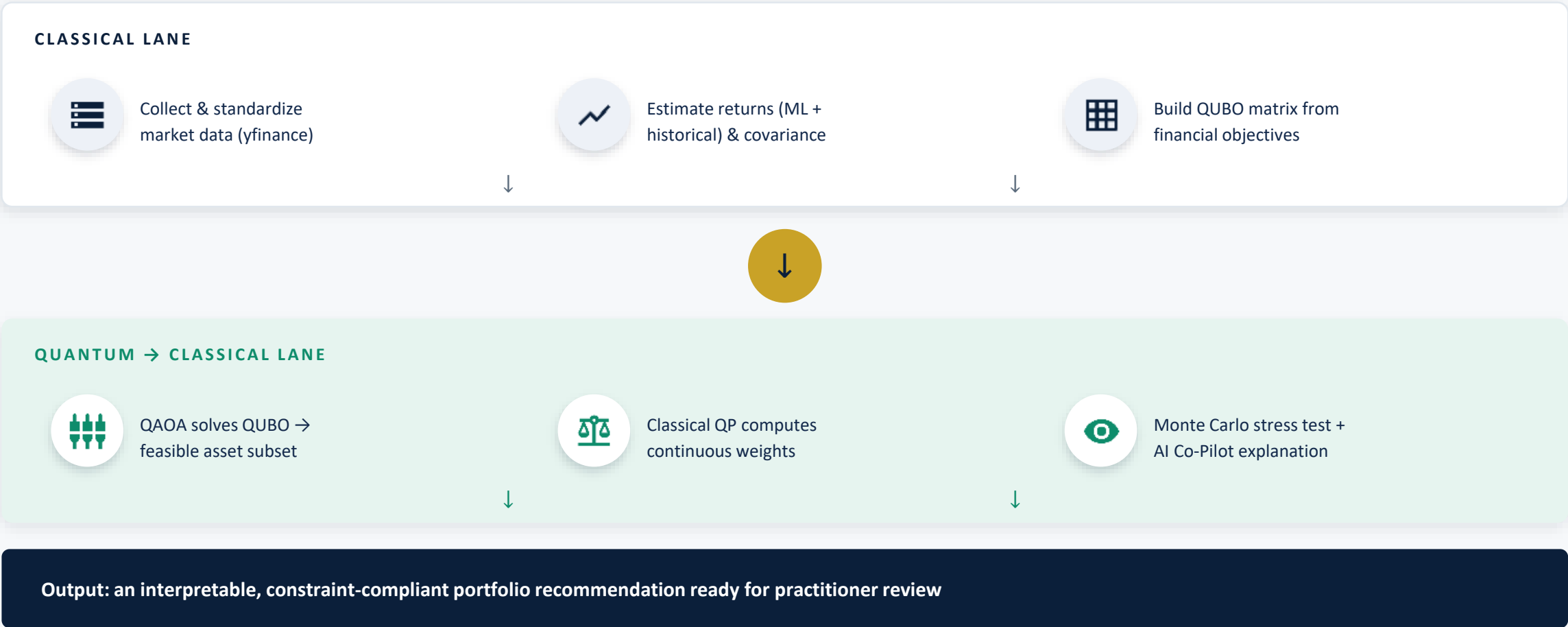
One pipeline, two computational paradigms — each stage handed to the technique best suited to solve it.



Only Stage 2 runs on quantum hardware or simulators. If unavailable, a classical fallback (simulated annealing / greedy heuristics) solves the identical QUBO objective — the downstream pipeline never changes.

Portfolio Optimization Workflow

From raw market data to an auditable investment recommendation — traced across the quantum and classical lanes.



Classical Portfolio Weight Optimization

With the asset subset fixed by QAOA, the remaining problem is a pure continuous convex program — efficiently solved by established QP solvers.

CONVEX OPTIMIZATION OBJECTIVE

minimize $w^T \Sigma w$

subject to:

$$\sum w_i = 1 \quad w_i \geq 0 \quad w_i \leq w_{\max}$$

Alternative objective: maximize the Sharpe ratio, balancing expected return against volatility instead of minimizing variance alone.

The classical benchmark portfolio follows this same solver — differing only in how candidate assets were screened (conventional vs. QAOA).

WHY THIS STAGE IS "EASY"



Discreteness already resolved

The hard combinatorial decision was solved upstream by QAOA



Pure quadratic program

Only continuous variables remain — classical QP solvers excel here



Fast, deterministic, mature

Decades of numerical optimization tooling apply directly

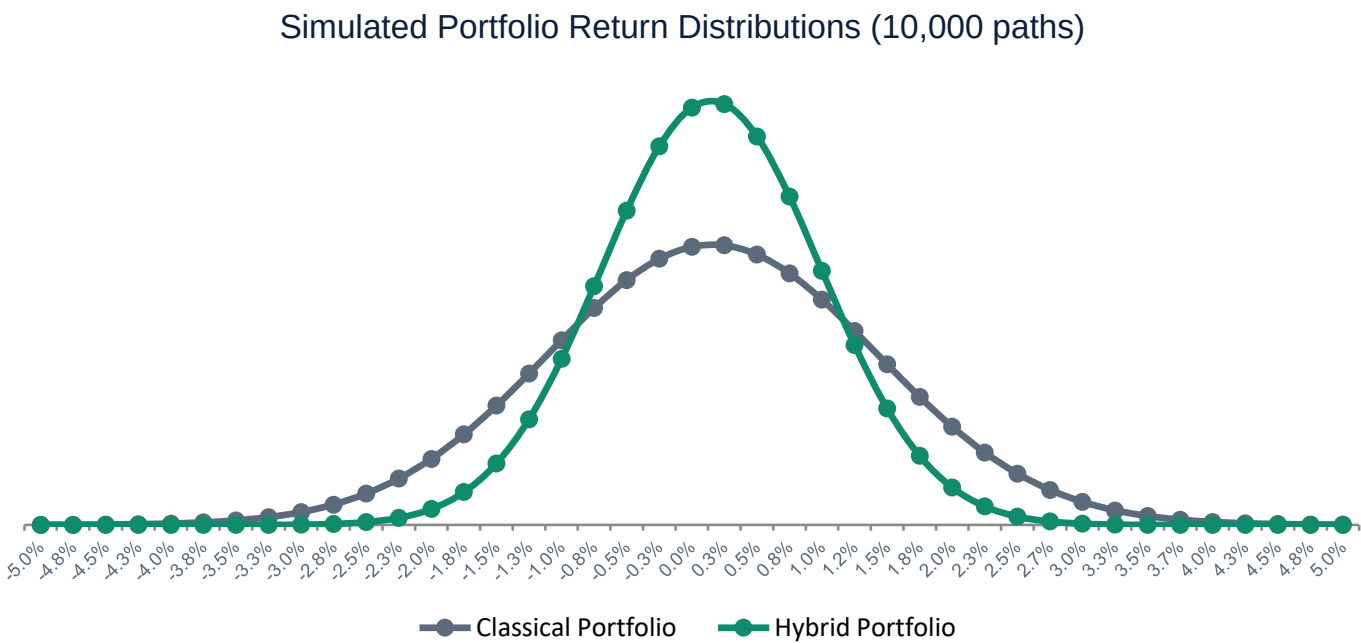


Isolates the comparison

Same weighting method for hybrid & classical benchmark → fair test

Monte Carlo Risk Analysis

Point estimates of return and variance hide downside risk. 10,000 simulated paths reveal the full distribution.



WHY IT MATTERS



10,000

Independent simulation paths, fixed seed



VaR (5%)

95% confidence downside-loss estimate



Normal Assumption

Efficient baseline; does not model fat tails

The hybrid portfolio's tighter, taller distribution reflects its ~34% lower volatility relative to the classical benchmark — a visibly narrower downside.

Explainable AI Portfolio Co-Pilot

Optimization outputs must be auditable. The Co-Pilot translates each selection decision into plain-language rationale.

EVALUATION CRITERIA PER ASSET

- ✓ Above-median predicted return
- ✓ High-liquidity threshold met
- ✓ Relatively low transaction cost
- ✓ Significant portfolio allocation weight

Assets meeting none of the above are labeled diversification components — included to reduce overall portfolio risk.



"Selected due to strong predicted return, high market liquidity, and significant portfolio allocation."



"Included primarily to improve portfolio diversification despite moderate expected return."

Plus a portfolio-level summary: objective, market conditions, expected return, risk, liquidity, cost, and hybrid-vs-classical rationale.

Experimental Setup

Both portfolios — hybrid and classical — were evaluated on identical real market data for a fair, apples-to-apples comparison.



Data Source

Real historical daily adjusted-close prices via Yahoo Finance (yfinance)



Asset Universe

Domestic equity, growth equity, value equity, government bonds, international bonds



Return Estimation

ML-based predictions blended with historical return series



Risk Modeling

Covariance matrix computed from historical returns to capture co-movement



Preprocessing

Missing-value removal, trading-date alignment, feature standardization



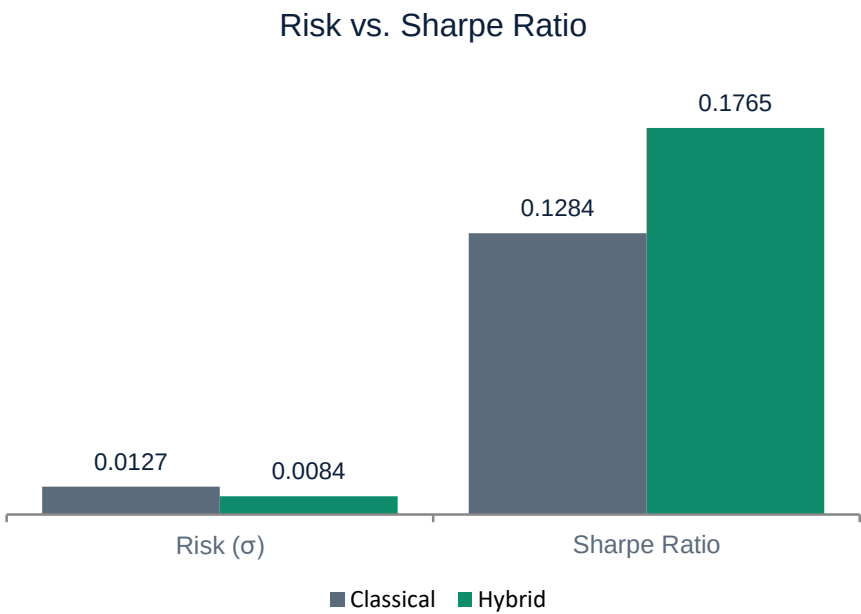
Fair Comparison

Hybrid and classical benchmark evaluated on identical market conditions

Performance Comparison

Table 1 — hybrid vs. classical portfolio performance across six evaluation metrics.

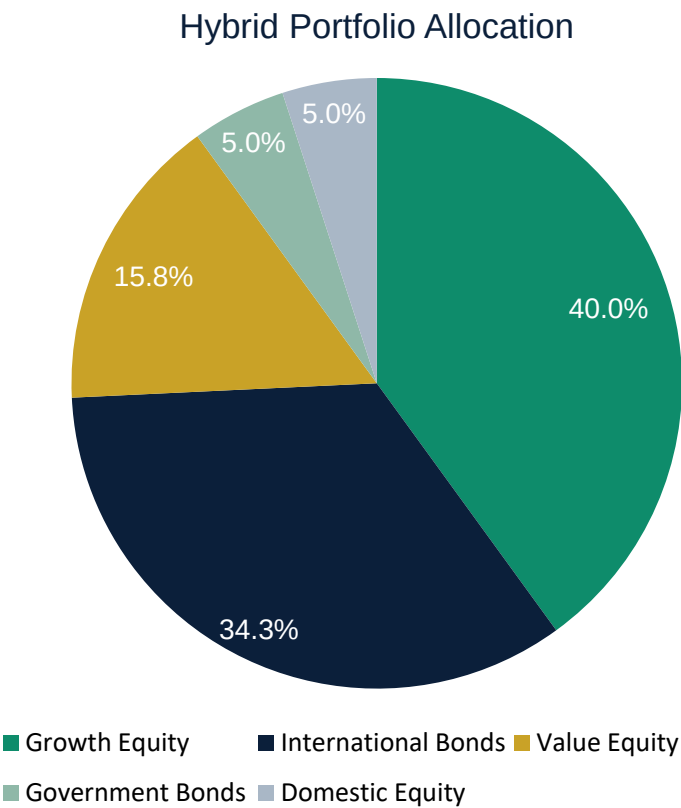
Metric	Classical	Hybrid	Change
Expected Return	0.0016	0.0015	−6.3%
Portfolio Risk (σ)	0.0127	0.0084	−34%
Sharpe Ratio	0.1284	0.1765	+37%
Avg. Liquidity	0.9695	0.9725	+0.3%
Avg. Transaction Cost	0.0016	0.0013	−19%
Assets Selected	—	5	K = 5



The hybrid portfolio trades a marginal 6.3% return reduction for a 34% risk cut — delivering a 37% higher Sharpe ratio and better operational efficiency.

Portfolio Composition Analysis

The QAOA-selected, classically-weighted hybrid portfolio — five assets spanning equity and fixed income.



	Growth Equity	40.0%
Largest allocation — strongest predicted return with acceptable liquidity & cost		
	International Bonds	34.25%
Second-largest — stability via low correlation with equities		
	Value Equity	15.75%
Enhances return while preserving high liquidity		
	Government Bonds	5.0%
Minimum allocation — diversification & stability		
	Domestic Equity	5.0%
Minimum allocation — broadens diversification base		

The optimizer diversified rather than concentrated — balancing return, correlation, liquidity, and transaction-cost objectives encoded in the QUBO.

Advantages Over Classical Portfolio Optimization



Native Discreteness

QAOA handles cardinality & selection directly
— no heuristic patchwork



Lower Portfolio Risk

~34% volatility reduction vs. the classical
benchmark



Higher Sharpe Ratio

0.1765 vs. 0.1284 — materially better risk-
adjusted return



Better Liquidity Profile

Average liquidity improves from 0.9695 to
0.9725



Lower Transaction Costs

Average cost falls from 0.0016 to 0.0013
(-19%)



Built-In Explainability

Every decision carries a human-readable
rationale



Zero Constraint Violations

Cardinality, liquidity, cost, and allocation rules
fully satisfied

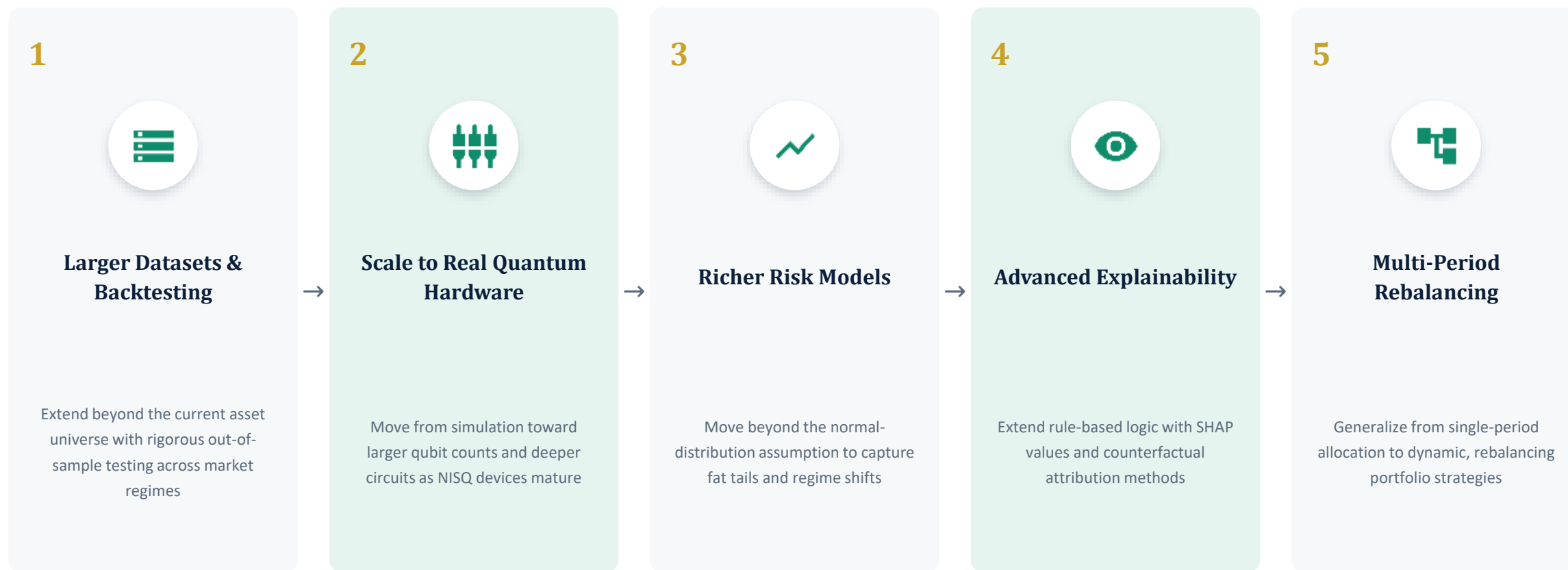


Scales With Complexity

Combinatorial search benefits grow as the
asset universe expands

Future Scope

Honest limitations today define a clear, actionable research roadmap for tomorrow.



Research Contributions

Five concrete contributions bridging quantum optimization and institutional portfolio management.

1



A QUBO-based formulation for constrained asset selection using QAOA

2



A hybrid optimization pipeline combining quantum selection with classical weighting

3



A Monte Carlo stress-testing framework for downside risk and Value-at-Risk

4



An explainable AI Portfolio Co-Pilot interpreting optimization decisions

5



A comparative evaluation of hybrid vs. classical portfolio construction

CONCLUSION

Toward Explainable Quantum Finance

This framework does not claim quantum advantage today. It proves something more durable: that quantum and classical methods, deployed where each is strongest, can already deliver measurably better, more transparent investment decisions — a practical pathway from NISQ-era hardware to institutional trust.

−34%

PORTFOLIO RISK

+37%

SHARPE RATIO

0

CONSTRAINT VIOLATIONS

The future of institutional finance is not quantum instead of classical — it is quantum with classical, and both explained.